

Holonomy and spectral bounds for signed cycle squares

Research manuscript increment 5 October 2026

Abstract

We study the adjacency spectral radius of real signings of the cycle square $C_n(1, 2)$. For the triangle-sign word $(+, +, -, +, -, -, +, -)$ repeated on $8m$ vertices, we give a self-contained proof of the exact radii in the two Hamilton-cycle holonomy sectors. Negative holonomy has radius $\sqrt{4 + \sqrt{8 + 2 \cos(\pi/m)} + \sqrt{26 - 6 \cos(\pi/m)}}$, strictly below the positive-holonomy value $\sqrt{4 + \sqrt{10 + 2\sqrt{5}}}$ for every finite m . This disproves the unrestricted-minimum conjecture in Suvagiya's September 2026 revision. Separately, an explicit signing on every $n \equiv 2 \pmod{8}$, $n \geq 50$, satisfies $\rho^2 < 198/25$. Its proof combines a rational finite entrance certificate with an analytic Riccati contraction and a uniform Schur-tail bound. Neither result determines the unrestricted minimum over all signings.

1 Introduction

For $n \geq 8$, the cycle square $C_n(1, 2)$ has vertex set $\mathbb{Z}/n\mathbb{Z}$ and edges $\{i, i+1\}$ and $\{i, i+2\}$. A signing assigns a value in $\{\pm 1\}$ to each edge. Its signed adjacency matrix A_σ is real symmetric, and we write

$$\rho(A_\sigma) = \max\{|\lambda| : \lambda \in \text{spec}(A_\sigma)\}, \quad \mu_n = \min_\sigma \rho(A_\sigma).$$

Both spectral edges matter in this minimization. An upper bound for $\lambda_{\max}$ alone does not give the required upper bound for ρ .

The earlier twisted-optimality conjecture [1] compared μ_n with the benchmark

$$\rho_{\text{tw}}(n)^2 = 4 + 2 \cos \frac{2\pi}{n} + 2 \cos \frac{4\pi}{n} \quad (n \text{ even}, n \geq 8). \quad (1)$$

That preprint has been withdrawn and incorporated into [2]. The revised Theorem 26 gives a period-eight signing with radius

$$r_* = \sqrt{4 + \sqrt{10 + 2\sqrt{5}}} = 2.7936044933 \dots \quad (2)$$

on $8m$ vertices for $m \geq 4$, and Conjecture 28 asserts $\mu_{8m} = r_*$ for $m \geq 4$. The construction has positive Hamilton-cycle holonomy; Remark 27 identifies the opposite holonomy as absent from the reported search. Changing that holonomy while preserving the triangle signs gives a strict improvement at every finite size.

Fix the word

$$t = (1, 1, -1, 1, -1, -1, 1, -1). \quad (3)$$

For $n = 8m$, begin with positive step-one edges and step-two signs $t_{i \bmod 8}$. Let A_m^+ be this adjacency matrix. Let A_m^- be obtained by reversing the signs of exactly three edges:

$$\{n-1, 0\}, \quad \{n-2, 0\}, \quad \{n-1, 1\}. \quad (4)$$

These are distinct edges for every $n \geq 8$.

Theorem 1.1. *For every integer $m \geq 1$, define*

$$R(s) = \sqrt{4 + \sqrt{8 + s + \sqrt{26 - 3s}}} \quad (-2 \leq s \leq 2).$$

Then

$$\rho(A_m^+) = R(2) = r_*, \quad \rho(A_m^-) = R\left(2 \cos \frac{\pi}{m}\right) < r_*. \quad (5)$$

Moreover, among all signings with the prescribed labeled triangle signs $t_{i \bmod 8}$, the negative-holonomy switching class is the unique minimizing switching class. In particular,

$$\mu_{8m} \leq R\left(2 \cos \frac{\pi}{m}\right) < r_* \quad (m \geq 4),$$

so Conjecture 28 of [2] is false as stated.

The uniqueness assertion is confined to the prescribed triangle signs. The theorem supplies no matching lower bound for arbitrary triangle words, and therefore no classification of unrestricted minimizers. The dispersion relation and exact finite formula are recorded in the earlier period-eight research manuscript [5]. The present application concerns the revised conjecture and makes explicit why its proposed endpoint is missed by every finite negative-holonomy grid.

A separate issue is to obtain a short proof for an explicit family outside the 8-divisible subsequence. The following statement is independent of Theorem 1.1.

Theorem 1.2. *Let $k \geq 6$, $n = 8k + 2$, and let B_n be the signed adjacency matrix with all step-one signs positive and the step-two sign word*

$$\tau = t^k \parallel (1, -1).$$

Then

$$\frac{198}{25} I_n - B_n^2 \succ 0, \quad \rho(B_n)^2 < \frac{198}{25} < \rho_{\text{tw}}(n)^2. \quad (6)$$

Appendix A gives the full analytic argument and its finite rational premises. The latter have been checked both through the block recurrence and independently from the signed graph by scalar Schur elimination. The theorem is thus a finite-certificate-assisted analytic result. The signing is inherited from the earlier residue-two construction; the increment is the analytic closure and its corrected exact certificates. The proof does not rely on the historical all-even classification.

The character interpretation of twists is part of established spectral methods. Luo and Roy [3], particularly their Theorems 3.6 and 3.13, study graph spectra over homology-character families and endpoint attainment. Their setting provides context for holonomy, but does not give an all-signing minimum for this fixed cycle-square support. The parity-closed-walk identity of Chen, van Dam, and Bu [4, Theorem 3.1] expresses signed moments averaged over all signings; an average is likewise not a uniform lower bound for each signing. Our proofs instead use the explicit fiber polynomial for one triangle word and exact positive-definiteness certificates for another.

2 Switching and finite holonomy

Write a_i for the sign of $\{i, i + 1\}$ and d_i for the sign of $\{i, i + 2\}$, with cyclic indices. The triangle sign and Hamilton-cycle holonomy are

$$\tau_i = a_i a_{i+1} d_i, \quad \alpha = \prod_{i=0}^{n-1} a_i.$$

Switching by vertex signs $g_i \in \{\pm 1\}$ replaces A by GAG , where $G = \text{diag}(g_i)$. It preserves the spectrum, every τ_i , and α .

Lemma 2.1. *For prescribed triangle signs $(\tau_i)_{i \in \mathbb{Z}/n\mathbb{Z}}$, there are exactly two switching classes, indexed by $\alpha \in \{\pm 1\}$. Each has a unique representative with*

$$a_0 = \cdots = a_{n-2} = 1, \quad a_{n-1} = \alpha, \quad d_i = \tau_i a_i a_{i+1}.$$

Proof. Set $g_0 = 1$ and recursively $g_{i+1} = g_i a_i$ for $0 \leq i < n-1$. After switching, the first $n-1$ step-one signs are positive. Their product with the remaining sign is α , so that remaining sign is α . The triangle equations force every step-two sign. Conversely, the displayed signs realize the prescribed data for either choice of α . A switch between two such representatives is constant along the path $0, 1, \dots, n-1$, hence constant on all vertices and acts trivially. Finally, distinct values of the switching invariant α cannot be switching equivalent. $\square$

For the repeated word (3), this gauge yields exactly A_m^+ and A_m^- in (4). Reversing only the Hamilton edge would change two triangle signs. The two additional step-two reversals are therefore essential.

Extend $u \in \mathbb{C}^{8m}$ to the integer line by

$$u_{i+8m} = \alpha u_i. \quad (7)$$

Let $t_i = t_{i \bmod 8}$. In either seam gauge the adjacency action is the restriction of

$$(Lu)_i = u_{i-1} + u_{i+1} + t_{i-2}u_{i-2} + t_i u_{i+2}. \quad (8)$$

In particular, the signs of all cut-crossing edges in the finite matrix agree with (7), including for $m = 1$.

Lemma 2.2. *The complexification of A_m^α is unitarily equivalent to*

$$\bigoplus_{z^m = \alpha} H(z),$$

where, in the order $0, 1, \dots, 7$,

$$H(z) = \begin{pmatrix} 0 & 1 & 1 & 0 & 0 & 0 & z^{-1} & z^{-1} \\ 1 & 0 & 1 & 1 & 0 & 0 & 0 & -z^{-1} \\ 1 & 1 & 0 & 1 & -1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & -1 & 1 & 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 1 & 1 & 0 & 1 & -1 \\ z & 0 & 0 & 0 & -1 & 1 & 0 & 1 \\ z & -z & 0 & 0 & 0 & -1 & 1 & 0 \end{pmatrix}. \quad (9)$$

Proof. For each root $z^m = \alpha$ and $v \in \mathbb{C}^8$, set $u_{8j+a} = m^{-1/2} z^j v_a$ for $0 \leq j < m$ and $0 \leq a < 8$. This satisfies (7). Substitution into (8) gives (9). For two different allowed phases z, w , the ratio $z\bar{w}$ is a nontrivial m th root of unity, so

$$\sum_{j=0}^{m-1} (z\bar{w})^j = 0.$$

Thus the m fiber spaces are mutually orthogonal, each has dimension eight, and together they exhaust dimension $8m$. The displayed map is unitary. Since $|z| = 1$, the fiber matrix is Hermitian. $\square$

3 The exact antiperiodic spectrum

Lemma 3.1. For $|z| = 1$ and $s = z + z^{-1}$,

$$\det(xI_8 - H(z)) = P(x, s) := x^8 - 16x^6 + 80x^4 - 128x^2 + 38 + s(-2x^4 + 16x^2 - 13) + s^2. \quad (10)$$

Its eight roots, with all signs independently chosen, are

$$\pm \sqrt{4 \pm \sqrt{8 + s \pm \sqrt{26 - 3s}}}. \quad (11)$$

They are all real, nonzero, and distinct for every $-2 \leq s \leq 2$.

Proof. Expanding the determinant of (9) as a Laurent polynomial in z gives

$$x^8 - 16x^6 + 80x^4 - 128x^2 + 40 + (z + z^{-1})(-2x^4 + 16x^2 - 13) + (z^2 + z^{-2}).$$

Using $z^2 + z^{-2} = s^2 - 2$ proves (10). This is a finite polynomial identity and can also be checked by the exact symbolic determinant in the accompanying verifier.

Put $y = x^2$. A translation gives

$$P(x, s) = (y - 4)^4 - (16 + 2s)(y - 4)^2 + s^2 + 19s + 38. \quad (12)$$

Solving this as a quadratic in $(y - 4)^2$ gives

$$u_{\pm}(s) = 8 + s \pm \sqrt{26 - 3s}.$$

The function $s^2 + 19s + 38$ is increasing on $[-2, 2]$ and has value 4 at -2 . Since $8 + s > 0$, the identity

$$(8 + s)^2 - (26 - 3s) = s^2 + 19s + 38 \geq 4$$

proves $u_- > 0$. Also $u_+ > u_-$, and

$$u'_+(s) = 1 - \frac{3}{2\sqrt{26 - 3s}} > 0, \quad u_+(s) \leq 10 + 2\sqrt{5} < 16.$$

Consequently

$$0 < 4 - \sqrt{u_+} < 4 - \sqrt{u_-} < 4 + \sqrt{u_-} < 4 + \sqrt{u_+}.$$

These are the four distinct positive roots in y . Taking their positive and negative square roots gives exactly (11). $\square$

Proof of Theorem 1.1. Lemma 3.1 implies

$$\rho(H(z)) = R(s), \quad s = z + z^{-1}.$$

The derivative computation in that lemma and strict monotonicity of the outer square roots show that R increases strictly on $[-2, 2]$. For positive holonomy, the grid $z^m = 1$ contains $z = 1$, so its maximal value of s is 2. For negative holonomy, the grid is

$$z_j = \exp\left(\frac{(2j+1)\pi i}{m}\right), \quad 0 \leq j < m,$$

and its maximal value of s is $2 \cos(\pi/m)$. Lemma 2.2 therefore proves both equalities in (5). For every finite $m \geq 1$, $\cos(\pi/m) < 1$, which proves the strict inequality. Lemma 2.1 shows that these are exactly the two switching classes for the prescribed triangle signs, so the strict comparison proves the stated constrained uniqueness. Since A_m^- is an admissible real signing, the unrestricted upper bound follows. $\square$

To identify (2) with the algebraic number used in the revised conjecture, put

$$f(x) = x^4 - 2x^3 - 6x^2 + 12x - 4.$$

An exact expansion gives $P(x, 2) = f(x)f(-x)$. On $x \geq \sqrt{6}$,

$$f''(x) = 12(x^2 - x - 1) > 0, \quad f'(\sqrt{6}) = 12\sqrt{6} - 24 > 0, \quad f(\sqrt{6}) = -4.$$

Hence f has exactly one root on that ray. For $x > \sqrt{6}$, $f(-x) - f(x) = 4x(x^2 - 6) > 0$. The largest positive root of $P(x, 2)$ is $R(2) > \sqrt{6}$. It must be a root of f , because $f(-R(2)) = 0$ would force $f(R(2)) < 0$ and then a still larger positive root of f , contrary to maximality. Thus $R(2)$ is the largest real root of f .

Corollary 3.2. *The negative-holonomy radii increase strictly to r_* as m increases, and*

$$r_* - \rho(A_m^-) = \frac{\pi^2}{4r_*\sqrt{10 + 2\sqrt{5}}} \left(1 - \frac{3}{4\sqrt{5}}\right) m^{-2} + O(m^{-4}). \quad (13)$$

Proof. The sequence $2 \cos(\pi/m)$ increases strictly to 2, and R is strictly increasing and smooth near 2. Since

$$2 \cos(\pi/m) = 2 - \pi^2 m^{-2} + O(m^{-4}), \quad R'(2) = \frac{1 - 3/(4\sqrt{5})}{4r_*\sqrt{10 + 2\sqrt{5}}},$$

Taylor's theorem proves (13). □

There is no parity exception: when m is odd, the allowed phase $z = -1$ has $s = -2$ and cannot recover the omitted maximum at $s = 2$. At $m = 1$ the single fiber is $H(-1)$, with $\rho(A_1^-) = \sqrt{6 + \sqrt{2}}$.

4 An exact certificate at order 32

At $m = 4$, Theorem 1.1 gives

$$\rho(A_4^-) = \sqrt{4 + \sqrt{8 + \sqrt{2} + \sqrt{26 - 3\sqrt{2}}}} = 2.7842697993 \dots \quad (14)$$

The characteristic polynomial computed directly from the signed 32×32 integer matrix is $Q(x)^2$, where

$$Q(x) = x^{16} - 32x^{14} + 416x^{12} - 2816x^{10} + 10568x^8 \\ - 21632x^6 + 22168x^4 - 9408x^2 + 1262.$$

This agrees with $P(x, \sqrt{2})P(x, -\sqrt{2})$; each value of s occurs twice on the antiperiodic four-cell grid.

A separate rational certificate avoids both the fiber calculation and numerical eigenvalues. Every leading principal minor of each integer matrix

$$279I_{32} - 100A_4^-, \quad 279I_{32} + 100A_4^-$$

is strictly positive. The certificate records all 64 integer minors; fraction-free elimination checks every division exactly. Sylvester's criterion yields $\rho(A_4^-) < 279/100$. Meanwhile,

$$f(279/100) = -\frac{6766519}{100000000} < 0, \quad f(14/5) = \frac{76}{625} > 0.$$

The monotonicity proved above gives the exact separation

$$\rho(A_4^-) < 2.79 < r_* < 2.8.$$

The direct graph certificate and the all- m analytic proof provide independent ways of checking the failure at $n = 32$.

The next extremal question remains to determine μ_{8m} and its minimizing switching classes. The period-eight negative-holonomy family is an explicit upper bound, not a proof of optimality. Establishing optimality would require a lower bound over arbitrary triangle words, including nonperiodic words. Similarly, Theorem 1.2 is an existence theorem for one signing on each order $8k + 2$. Combining such witnesses does not certify a complete truth set for the twisted benchmark without independent lower bounds at the complementary orders.

A A uniform residue-two Schur bound

We prove Theorem 1.2. In this appendix $n = 8k + 2 = 4\ell + 2$, where $\ell = 2k$ is even and $k \geq 6$. The letter ℓ counts four-vertex blocks and is distinct from the cell count m in the main text. Put

$$M_n = \frac{198}{25}I_n - B_n^2.$$

All matrix inequalities in this appendix concern real symmetric matrices; $X \succ 0$ means positive definite. The norm $\|\cdot\|_2$ is the Euclidean operator norm, and $\|\cdot\|_F$ is the Frobenius norm.

A.1 The exact block recurrence

Partition the vertices into

$$V_0 = \{0, 1\}, \quad V_j = \{2 + 4(j-1), \dots, 5 + 4(j-1)\} \quad (1 \leq j \leq \ell).$$

The diagonal block on each V_j , $j \geq 1$, is

$$D = \begin{pmatrix} 98/25 & 0 & -1 & 0 \\ 0 & 98/25 & 0 & -1 \\ -1 & 0 & 98/25 & 0 \\ 0 & -1 & 0 & 98/25 \end{pmatrix}.$$

The block $M_n[V_j, V_{j+1}]$ equals E_+ for odd j and E_- for even j , where

$$E_+ = \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ -1 & 2 & 1 & 0 \\ 2 & -1 & 0 & -1 \end{pmatrix}, \quad E_- = \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ -1 & -2 & 1 & 0 \\ -2 & -1 & 0 & -1 \end{pmatrix}.$$

The exceptional blocks are

$$\begin{aligned} G_0 &= M_n[V_0, V_0] = (98/25)I_2, & H_0 &= D, \\ R_0 &= M_n[V_0, V_1] = \begin{pmatrix} -1 & -2 & 1 & 0 \\ -2 & -1 & 0 & -1 \end{pmatrix}, \\ C_0 &= M_n[V_0, V_\ell] = \begin{pmatrix} -1 & 0 & -1 & 0 \\ 0 & -1 & 0 & -1 \end{pmatrix}, & W_0 &= M_n[V_1, V_\ell] = \begin{pmatrix} 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}. \end{aligned}$$

All other nonadjacent bulk blocks vanish. To verify these identities at all lengths, the only nonzero entries of B_n^2 on the forward cyclic diagonals, apart from transposes, are

$$\begin{aligned} (B_n^2)_{i,i} &= 4, & (B_n^2)_{i,i+1} &= \tau_{i-1} + \tau_i, & (B_n^2)_{i,i+2} &= 1, \\ (B_n^2)_{i,i+3} &= \tau_i + \tau_{i+1}, & (B_n^2)_{i,i+4} &= \tau_i \tau_{i+2}. \end{aligned}$$

They follow by enumerating the possible two-edge walks. Since $n \geq 50$, these cyclic offsets do not collide. Substitution of $t^k \|(1, -1)$ gives the displayed blocks without a length-dependent assumption.

Let $E_j = E_+$ for even j and $E_j = E_-$ for odd j . Set $X_0 = D$. Before closing the right boundary, elimination of the current four-vertex pivot gives the length-independent recurrence

$$\begin{aligned} X_{j+1} &= D - E_j^T X_j^{-1} E_j, & R_{j+1} &= -R_j X_j^{-1} E_j, \\ W_{j+1} &= -E_j^T X_j^{-1} W_j, & G_{j+1} &= G_j - R_j X_j^{-1} R_j^T, \\ H_{j+1} &= H_j - W_j^T X_j^{-1} W_j, & C_{j+1} &= C_j - R_j X_j^{-1} W_j. \end{aligned} \quad (15)$$

The terminal elimination index is $p = \ell - 2$, which is even. At that step only, the coupling to the retained last block becomes $W_p + E_+$. Thus the normalized six-dimensional core is

$$S_\ell = \begin{pmatrix} G_p - R_p X_p^{-1} R_p^T & C_p - R_p X_p^{-1} (W_p + E_+) \\ * & H_p - (W_p + E_+)^T X_p^{-1} (W_p + E_+) \end{pmatrix}. \quad (16)$$

Here $*$ denotes the transpose block. Repeated Schur congruence proves that $M_n \succ 0$ if all eliminated pivots and S_ℓ are positive. The subscript of S_ℓ is a block count: in particular, S_{26} corresponds to graph order 106.

A.2 Finite exact premises

Define $F_\pm(X) = D - E_\pm^T X^{-1} E_\pm$ and $\Phi = F_- \circ F_+$. Put $Z = \Phi^{12}(D) = X_{24}$, $Y = F_+(Z)$, and $L_Z = Z^{-1} E_+ Y^{-1} E_-$. Use the rational matrices

$$\begin{aligned} P &= \frac{1}{10000} \begin{pmatrix} 10766 & 87 & 19 & 974 \\ 87 & 12664 & 148 & -2418 \\ 19 & 148 & 10093 & -25 \\ 974 & -2418 & -25 & 14009 \end{pmatrix}, \\ Q &= \frac{1}{10000} \begin{pmatrix} 11503 & 614 & 990 & -1101 \\ 614 & 10470 & 15 & 113 \\ 990 & 15 & 12299 & -2632 \\ -1101 & 113 & -2632 & 13260 \end{pmatrix}. \end{aligned}$$

Set $r = 10^{-10}$. The following finite inequalities are the complete certificate premises used below:

1. $X_0, \dots, X_{24} \succ 0$ and $Z, Y \succ I_4/2$;
2. $(9/10)I_4 \prec P, Q \prec 2I_4$;
3. $L_Z^T P L_Z \prec (2/5)P$ and $L_Z Q L_Z^T \prec (2/5)Q$;
4. $\|\Phi(Z) - Z\|_F < r/40$;
5. $R_{24} Q R_{24}^T \prec 10^{-10}I_2$ and $W_{24}^T Q W_{24} \prec 10^{-10}I_4$;
6. $S_{26} \succ (1/50)I_6$;

7. $S_\ell \succ 0$ for $\ell = 12, 14, 16, 18, 20, 22, 24$.

They have been verified using exact rational arithmetic. For clarity, their verification requires no limiting or approximate matrix: all matrices are specified by the displayed rational initial data and a finite number of recurrence steps. Positive definiteness is checked by successive scalar Schur pivots. For a symmetric matrix K , start with $K^{(0)} = K$; at each step require $d_j = K_{11}^{(j)} > 0$ and replace the remaining block by $K^{(j+1)} = K_{22}^{(j)} - K_{21}^{(j)}(d_j)^{-1}K_{12}^{(j)}$. All resulting scalars are rational, so each test is exact. The residual test in item 4 is the rational inequality

$$\sum_{a,b} (\Phi(Z) - Z)_{ab}^2 < (r/40)^2.$$

The accompanying certificate records the matrices and the positive pivots. An independent replay constructs B_{106} from its edges, eliminates 96 scalar vertices to recover the entrance data, and verifies items 1–5 without generating that entrance by the block recurrence. Direct scalar elimination from the graphs also verifies the seven base orders 50, 58, 66, 74, 82, 90, 98 and the normalized seed at 106.

A.3 A local Riccati contraction

For symmetric U , define the order-unit norm

$$|U|_P = \inf\{a \geq 0 : -aP \preceq U \preceq aP\} = \|P^{-1/2}UP^{-1/2}\|_2.$$

Consider the closed ball $\mathcal{K} = \{X = X^T : |X - Z|_P \leq r\}$. Since $P \prec 2I$,

$$\|X - Z\|_2 \leq 2r =: e.$$

The certified lower bound $Z \succ I/2$ gives $X \succ I/3$ and $\|X^{-1}\|_2 \leq 3$. The inverse identity yields

$$\|X^{-1} - Z^{-1}\|_2 \leq 6e.$$

Both $E_\pm$ have squared Frobenius norm 14. Writing $Y_X = F_+(X)$, we obtain

$$\|Y_X - Y\|_2 \leq 84e < 1/6, \quad Y_X \succ I/3, \quad \|Y_X^{-1} - Y^{-1}\|_2 \leq 504e.$$

Define $L(X) = X^{-1}E_+Y_X^{-1}E_-$. Splitting its difference from L_Z into two terms gives

$$\|L(X) - L_Z\|_2 \leq 14(6 \cdot 3 + 2 \cdot 504)e = 14364e. \quad (17)$$

For the two transfer norms

$$\|L\|_{P,\text{col}} = \|P^{1/2}LP^{-1/2}\|_2, \quad \|L\|_{Q,\text{row}} = \|Q^{-1/2}LQ^{1/2}\|_2,$$

conversion from the Euclidean norm costs at most $\sqrt{20/9} < 3/2$. Therefore the perturbation in either norm is less than $21546e = 43092r < 10^{-4}$. The certified bounds at Z , together with $\sqrt{2/5} + 10^{-4} < 2/3$, imply throughout $\mathcal{K}$ that

$$L(X)^T P L(X) \prec \frac{4}{9}P, \quad L(X) Q L(X)^T \prec \frac{4}{9}Q. \quad (18)$$

Direct differentiation gives $D\Phi_X[U] = L(X)^T U L(X)$. If $-aP \preceq U \preceq aP$, the first inequality in (18) bounds this derivative in $|\cdot|_P$ by $4a/9$. Integrating along a line segment in the convex ball proves that Φ is $4/9$ -Lipschitz there.

The residual premise gives $|\Phi(Z) - Z|_P < (10/9)(r/40) = r/36$. Consequently

$$|\Phi(X) - Z|_P < r/36 + (4/9)r = (17/36)r < r.$$

The ball is invariant. Banach's fixed-point theorem supplies a unique $X_* \in \mathcal{K}$ with $\Phi(X_*) = X_*$. The actual even trajectory starts at Z , so, with $\theta = 4/9$,

$$\|X_{24+2h} - X_*\|_2 \leq 4r\theta^h \quad (h \geq 0). \quad (19)$$

The constant $4r$ follows already from the diameter of the ball. All these even pivots and their intermediate odd partners exceed $I/3$. Together with the finite entrance this proves positivity of every pivot needed at any length.

A.4 Response decay

Two consecutive updates in (15) act by $R \mapsto RL(X)$ and $W \mapsto L(X)^T W$. For these two orientations, respectively, use

$$\alpha(R) = \|RQ^{1/2}\|_2, \quad \beta(W) = \|Q^{1/2}W\|_2.$$

The second inequality in (18) contracts both quantities by $q = 2/3$. Hence the entrance premises imply

$$\begin{aligned} R_{24+2h} Q R_{24+2h}^T &\prec 10^{-10} q^{2h} I_2, \\ W_{24+2h}^T Q W_{24+2h} &\prec 10^{-10} q^{2h} I_4. \end{aligned}$$

Since $Q \succ (9/10)I$, it follows that

$$\|R_{24+2h}\|_2, \|W_{24+2h}\|_2 < a q^h, \quad a = 1/30000. \quad (20)$$

Each one-step transfer has norm at most $3\sqrt{14} < 12$. Thus both members of the h th pair obey

$$\|R_j\|_2, \|W_j\|_2 \leq b q^h, \quad b = 12a = 1/2500, \quad j \in \{24 + 2h, 25 + 2h\}. \quad (21)$$

The row orientation LQL^T is needed here. A bound on $L^T QL$ alone would not justify these response estimates.

A.5 The limiting core and its uniform tail

Use the actual infinite trajectory, not a trajectory with constant pivots, to define the absolutely convergent series

$$\begin{aligned} G_\infty &= G_0 - \sum_{j \geq 0} R_j X_j^{-1} R_j^T, \\ H_\infty &= H_0 - \sum_{j \geq 0} W_j^T X_j^{-1} W_j, \\ C_\infty &= C_0 - \sum_{j \geq 0} R_j X_j^{-1} W_j. \end{aligned}$$

Convergence follows from (21) and $\|X_j^{-1}\|_2 \leq 3$ after $j = 24$. Set

$$S_\infty = \begin{pmatrix} G_\infty & C_\infty \\ C_\infty^T & H_\infty - E_+^T X_*^{-1} E_+ \end{pmatrix}.$$

Take even $\ell \geq 26$ and write its terminal index as $p = \ell - 2 = 24 + 2h$. Each of the three Schur-series suffixes after index p is bounded in norm by

$$T_h = \frac{6b^2q^{2h}}{1 - q^2}. \quad (22)$$

Indeed, every increment in the j th pair has norm at most $3b^2q^{2j}$, and there are two increments per pair. The bound in (22) harmlessly includes the term at p as well.

The terminal linear terms from (16) satisfy

$$\begin{aligned} \|R_p X_p^{-1} E_+\|_2 &\leq 12aq^h, \\ \|W_p^T X_p^{-1} E_+ + E_+^T X_p^{-1} W_p\|_2 &\leq 24aq^h. \end{aligned}$$

The inverse identity and (19), with $\delta = 4r$, give

$$\|X_p^{-1} - X_*^{-1}\|_2 \leq 9\delta\theta^h, \quad \|E_+^T(X_p^{-1} - X_*^{-1})E_+\|_2 \leq 144\delta\theta^h.$$

For the differences of the three blocks in $S_\ell - S_\infty$ we therefore have

$$\begin{aligned} \|\Delta G\|_2 &\leq T_h, \\ \|\Delta C\|_2 &\leq T_h + 12aq^h, \\ \|\Delta H\|_2 &\leq T_h + 24aq^h + 144\delta\theta^h. \end{aligned}$$

The block estimate

$$\left\| \begin{pmatrix} U & V \\ V^T & Z \end{pmatrix} \right\|_2 \leq \|U\|_2 + 2\|V\|_2 + \|Z\|_2$$

now gives the explicit uniform tail bound

$$\begin{aligned} \|S_\ell - S_\infty\|_2 &\leq 4T_h + 48aq^h + 144\delta\theta^h \\ &\leq \frac{251089}{156250000} < \frac{1}{500}. \end{aligned} \quad (23)$$

The rational middle expression is the value of this majorant at $h = 0$; each geometric factor decreases for $h \geq 0$. Because the series were defined along the actual trajectory, the common initial terms cancel exactly. Only the terminal pure-pivot term requires the inverse-limit comparison above.

A.6 Completion of the spectral estimate

The exact seed $S_{26} \succ I_6/50$ and two applications of (23) imply, for every even $\ell \geq 26$,

$$\|S_\ell - S_{26}\|_2 < 2/500 = 1/250, \quad S_\ell \succ \left(\frac{1}{50} - \frac{1}{250} \right) I_6 = \frac{2}{125} I_6.$$

Both tail errors are required: the seed is a finite core, not the limiting core. The remaining even block counts $12 \leq \ell < 26$ are exactly the seven certified bases. Thus every core and every eliminated pivot is positive. Schur congruence proves $M_n \succ 0$, and the spectral theorem then gives $\rho(B_n)^2 < 198/25$.

This use of Schur congruence asserts positivity only. The lower bound $2/125$ for the reduced core is not a lower bound for the least Euclidean eigenvalue of the original matrix M_n . Finally, $\cos x > 1 - x^2/2$ for $x > 0$ and $\pi^2 < 10$ yield

$$\rho_{\text{tw}}(n)^2 > 8 - \frac{20\pi^2}{n^2} > 8 - \frac{200}{n^2} \geq \frac{198}{25} \quad (n \geq 50).$$

This completes the proof of Theorem 1.2.

Reproducibility. The accompanying source package includes `certificates/analytic/verify_r2_certificate.py` and `r2_exact_certificate.json`, together with the independent `replay_direct_graph_seed.py` and `replay_local_from_direct_graph.py` under `certificates/audit/`. For the separate antiperiodic check, the script `verify_antiperiodic_counterexample.py` under `certificates/new_conjecture/` constructs the graph, checks the symbolic identities, and verifies all recorded integer minors. The package README lists the exact execution commands and dependencies.

References

- [1] V. Suvagiya, *Signed circulants at the Ramanujan bound*, arXiv:2607.18334v1, 19 July 2026; withdrawn in version 2, 22 September 2026, and merged into [2]. <https://arxiv.org/abs/2607.18334>.
- [2] V. Suvagiya, *Parity families and signed spectra: kernel averaging, near-Ramanujan bounds, and exact circulant models*, arXiv:2607.17343v2, 22 September 2026, Theorem 26, Remark 27, and Conjecture 28. <https://arxiv.org/abs/2607.17343v2>.
- [3] Y. Luo and A. Roy, *Homological spectral graph theory and weighted cycle counting*, arXiv:2403.01550v4, 4 August 2026. <https://arxiv.org/abs/2403.01550v4>.
- [4] L. Chen, E. R. van Dam, and C. Bu, Spectra of power hypergraphs and signed graphs via parity-closed walks, *Journal of Combinatorial Theory, Series A* **207** (2024), 105909. <https://doi.org/10.1016/j.jcta.2024.105909>.
- [5] *The exact period-eight phase*, Section 4 of the earlier period-eight research manuscript, frozen repository source at commit 085ea698475b7b32e0ae57457ec903a922248f69. Archived source section.